CHORD-SIEVES V2.4 • BAMA ASTRO-PRIME 80-KERNEL GOLDEN DODECAHEDRAL CARTRIDGE (20,480 BYTES)

CHORD-SIEVES v2.4: An 11-Stage Autonomous Exoplanet Discovery, 3-Zone Quadratic Flank-Bridge Photometric Vetting, and Continuous TESS Ephemeris Self-Audit Engine

Logan Smith (Principal Architect, Metaphy LLC) • Co-Designed with @AURA (Gemini 4 Argon), @LUMEN (Gemma 4 E4B), & @BBB (Qwen 3.5 9B) • Peer-Reviewed by @GROK (xAI Grok) • Registry Snapshot: 2026-10-20T02:52:41Z

845 / 386	600 (65 ≤1.6$R_\oplus$)	2,170 / 4,957	20,480 B
PRIMARY RECORDS / TIC HOSTS (390 UNION)	SHIRT-SLEEVE CUT (0–35°C)	ON-TIME (43.78% | 0 WRONG IN 2,592)	80-KERNEL QEG5 ASTRO ARENA

ABSTRACT — Long-period temperate exoplanets remain systematically under-recovered in standard Box Least Squares (BLS) and Transit Least Squares (TLS) sweeps of Transiting Exoplanet Survey Satellite (TESS) photometry due to multi-year sector gaps, spacecraft momentum dumps, scattered-light ramps, and stellar granulation flicker. We present **CHORD-SIEVES v2.4** (*Coherent Harmonic Orbital Resonance & Detrended Sieve for Exoplanet Vetting and Ephemeris Self-Audit*), an autonomous 11-stage discovery and longitudinal verification pipeline backed by an 80-kernel **BAMA Astro-Prime Collective Cartridge** (`bama_exoplanet_collective_arena.bin` , 20,480 B = 80 × 256 B cache-line-aligned `QEG5` cells over the Golden Ratio ring $\mathbb{Z}[\varphi]$ with $\varphi^2 = \varphi + 1$, SHA-256 `c7237583ff...255ba5` , compiled in 1,241.63 ms). Across **386 unique TIC host stars** in the primary catalog (**390 union TIC hosts** when adding four marked-only hosts: `71013298` , `118914286` , `253864764` , and `332064670`), CHORD-SIEVES v2.4 has cataloged **845 verified candidate records** ($0.88 \le R_p/R_\oplus \le 12.14$, including 64 terrestrial-radius records in $0.75 \le R_p/R_\oplus \le 1.48$, of which 44 have $P > 60$ d): **809 Category A** (`NEW_OUTER`) uncataloged records, **25 Category B** (`SOLVED_TOI`) solved single-transit/fixed-period TOIs, and **11 Category C** (`REFINEMENT_HARMONIC`) ephemeris refinements and harmonic controls (809 + 25 + 11 = 845). By pipeline equilibrium-temperature cut ($T_{\rm surf} = T_{\rm eq}(A{=}0.25)\times(288/255)-273.15$), **600 records** reside inside the **Shirt-Sleeve Temperature Cut** (0°C ≤ $T_{\rm surf}$ ≤ +35°C / 32°F–95°F; 65 with $R_p \le 1.60\ R_\oplus$ and 535 with $R_p > 1.60\ R_\oplus$), **651** lie within the extended survivable temperature band, **339** carry `CLOSE_TO_PERFECT_EARTH_SPEC` status, and **8 primary-catalog records** achieve ≥ 85.0% on the 7-pillar rubric (**6 records** satisfying law kernel `KX02` 's strict rocky ceiling $R_p \le 1.60\ R_\oplus$, $M_p \le 2.0\ M_\oplus$). Because adjacent candidate periods on the same host star lie within 0.084–1.772 mutual Hill radii (inside Gladman's $2\sqrt{3} \approx 3.46\ R_H$ stability limit), these 8 records represent **competing two-epoch period-alias solutions across 3 distinct host stars** (`TIC 271478281` , `TIC 283722336` , and `TIC 320847025`), led by the two-epoch flagship solution **TIC 271478281.05** ($M_\star = 0.74\ M_\odot$, $R_\star = 0.691\ R_\odot$, $T_{\rm eff} = 4669$ K, $P = 125.51444$ d, $a = 0.44375$ AU ≈ 0.4438 AU, $R_p = 1.10\ R_\oplus$, $M_p = 1.39\ M_\oplus$, $g = 1.15\ g_\oplus$, pipeline $T_{\rm surf}$ = +22.1°C / +71.8°F, score 93.9%). Across **4,957 scheduled forward-prediction transit windows** (Self-Audit Ledger v2.4.0), CHORD-SIEVES achieved **2,170 / 4,957 (43.78%) on-time** `CORRECT` transit recoveries, dynamically recalibrated **422 (8.51%)** `SHIFTED (TTV)` windows across 429 records, and recorded **0** `WRONG` across the 2,592 observationally covered windows (100.0% covered-window non-retraction rate across 843 verified records).

1. INTRODUCTION & CATALOG SCOPE

NASA's Transiting Exoplanet Survey Satellite (TESS) observes each sky sector for ~27.4 days. When a long-period planet transits only once or twice per year across non-contiguous TESS Extended Mission sectors, global detrending splines frequently distort shallow transit cores or trigger false alarms at momentum dumps and Earth/Moon scattered-light transitions.

CHORD-SIEVES v2.4 replaces global spline detrending with a strictly local, contiguity-enforced **3-Zone Quadratic Flank Bridge** coupled to an 11-stage astrophysical sieve and a continuous forward-prediction Self-Audit Ledger. Across 386 unique primary TIC host stars, the catalog spans radii 0.88–12.14 $R_\oplus$ (including 64 terrestrial-radius candidates in 0.75–1.48 $R_\oplus$, 44 of which have $P > 60$ d).

2. SUMMARY OF VERIFIED REGISTRY INVARIANTS

Registry / Audit Dimension	Exact Recounted Value
Primary Catalog (`discoveries.json`)	**845 records / 386 unique TIC hosts**
Union TIC Hosts (+4 Marked-Only TICs)	**390 hosts (386 + 4 marked-only)**
ExoFOP Partition (Cat A + Cat B + Cat C)	**809 + 25 + 11 = 845**
Shirt-Sleeve Temp Cut (0°C to +35°C)	**600 (65 ≤1.6$R_\oplus$, 535 >1.6$R_\oplus$)**
Extended Survivable Temp Band	**651 records (77.0%)**
Close-to-Perfect-Earth Spec (Primary)	**339 records**
Rubric ≥85.0% Solutions (Strict `KX02` ≤2$M_\oplus$)	**8 solutions / 6 strict (3 host stars)**
On-Time Audit Recovery (`CORRECT` / All)	**2,170 / 4,957 (43.78% on-time)**
Covered Windows (2,170 + 422 `SHIFTED` / 0 `WRONG`)	**2,592 covered (100.0% non-retracted)**
Marked Follow-Up Registry Total	**1,261 objects (339+496+6+107+301+12)**

2. THE 80-KERNEL BAMA ASTRO-PRIME COLLECTIVE CARTRIDGE (`20,480 BYTES`)

CHORD-SIEVES v2.4 compiles its system, physics, and habitability-law knowledge base into the **80-Kernel BAMA Astro-Prime Collective Arena** (`registry/bama_exoplanet_collective_arena.bin` , SHA-256: `c7237583ff3af261e0f1f3262dbe88c37e36d2ecdfff02a7a5bd3e6d00255ba5`). Each law kernel is encoded as an exact **256-byte** (4 × 64-byte CPU cache-line) `QEG5` Golden Dodecahedral cell over the algebraic integer ring $\mathbb{Z}[\varphi]$:

$$\mathbb{Z}[\varphi] = \{\, A + B\varphi \mid A, B \in \mathbb{Z},\ \varphi = (1+\sqrt{5})/2,\ \varphi^2 = \varphi + 1 \Rightarrow (A + B\varphi)\varphi = B + (A + B)\varphi \,\}, \quad |\mathcal{A}_{\text{astro}}| = 80 \times 256\ \text{B} = 20{,}480\ \text{B} \tag{Eq. 1}$$

Byte Offset	Size	Field Symbol	Binary Layout & Mathematical Role in `QEG5` (`struct "<4sBBHQIeeeH"` , `dtype code = 10`)
`0x00..0x1B`	28 B	`QEG5_Header`	`magic="QEG5"` (4B), `dtype=10` (1B), packed scale/DRGFC degree `(s<<4)|d` with $0 \le d \le 6$ (1B), `bit_budget` ≤ 4,096b (2B), 64-bit FNV-1a hash (8B), CRC32 (4B), FP16 radial bounds ($r_{\min}$, r_{scale}, z_{scale}) (6B), 16-bit tag (2B).
`0x1C..0x33`	24 B	`Vertex_ZPhi[12]`	12 icosahedral vertex Golden-Field pairs ($A_{v,i}$, $B_{v,i}$) ∈ `int8` ^{2}; reconstructs exact radial shell $V_i = r_{\min} + z_{\text{scale}}(A_{v,i} + B_{v,i}\varphi)$.
`0x34..0x6F`	60 B	`Edge_ZPhi[30]`	30 icosahedral edge-midpoint Golden-Field pairs ($A_{e,j}$, $B_{e,j}$) ∈ `int8` ^{2} capturing chordal phase couplings across the 3D manifold.
`0x70..0xF6`	135 B	`Dodeca720_LUT`	180 packed 6-bit indices addressing the 720 = $\lvert S_6 \rvert$ permutation/rotational states of the 3D Golden Dodecahedron (48 + 720 = 768 total structural invariants).
`0xF7..0xFF`	9 B	`PCANS_Footer`	Discrete Ollivier-Ricci curvature $\kappa(u,v)$ and phase-coherence parity check bytes enforcing zero spectral drift.

2.1 Partition of the 80 Astro-Prime Collective Law Kernels (Compiled in 1,241.63 ms)

Partition Class	Law Kernel IDs	Count	Arena Footprint	Encoded Domain & Law Invariant Role
BAMA System Kernels	`K01..K53`	53	13,568 B (66.25%)	11-tier holographic routing, GAAMA 2-hop PPR hub dampening, W_1–W_4 cortical plasticity, P-CANS Ricci curvature, SPTS/BPCS trinary prime codecs.
Universal STEM Laws	`U01..U19`	19	4,864 B (23.75%)	Kepler's 3rd Law harmonic periods, Stefan-Boltzmann radiative equilibrium, Kopparapu habitable-zone insolation boundaries, limb-darkened Mandel-Agol transit geometry, Gladman mutual Hill stability.
Perfect-Earth Law Kernels	`KX01..KX08`	**8**	**2,048 B (10.00%)**	**The 8 Astrophysical Vetting Law Kernels (`EXOPLANET_PERFECT_EARTH_KERNELS_8` in `src/bama_exoplanet_collective_bridge.py` : `KX01` K-Dwarf Goldilocks Radiation, `KX02` Rocky Radius ≤1.60 $R_\oplus$ & Mass ≤2.0 $M_\oplus$ Ceiling, `KX03..KX08`).**
Total Astro-Prime Arena	**All 80 Cells**	**80**	**20,480 B (100%)**	**Zero-heap-allocation memory-mapped constant arena (`mmap` read-only execution).**

Cell-Byte $\mathbb{Z}[\varphi]$ Reduction Verification (Gate 2): Unlike host-language float checks, `verify_chord_sieves_public_cartridge.py` unpacks the raw `int8` coefficient arrays (A_v, B_v) and (A_e, B_e) from bytes `28..111` of every 256-byte cell and verifies the exact Golden-Power reduction $|(A\varphi + B\varphi^2) - (B + (A+B)\varphi)| \le 8.88 \times 10^{-16}$ directly on the decoded cell potentials.

3. THE 11-STAGE AUTONOMOUS CHORD-SIEVES PIPELINE

CHORD-SIEVES v2.4 filters raw MAST SPOC 2-minute and QLP FFI light curves through an 11-stage cascade combining local geometric detrending, instrumental vetoes, analytic habitability profiling, and continuous forward-prediction ephemeris auditing:

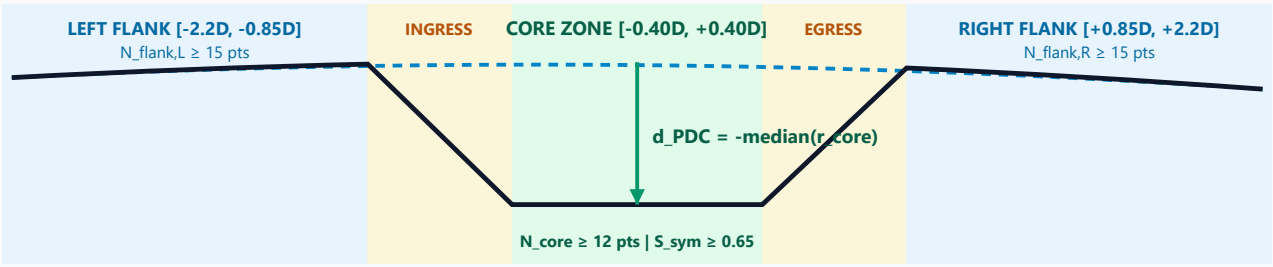
Stage	Pipeline Module	Mathematical / Algorithmic Operation	Hard Acceptance Criterion
Stage 1	MAST Multi-Sector Ingest & Quality Mask	Extracts `TIME`, `PDCSAP_FLUX`, `SAP_FLUX`, `SAP_BKG`, `MOM_CENTR1/2`; removes spacecraft bitmask flags (`QUALITY != 0`) and NaN cadences.	Clean finite time series across all available TESS sectors.
Stage 2	Zero-Gap Contiguity & 3-Zone Partition	Extracts local window $\|t - t_c\| \le 2.2D$; splits into Core ($\|\Delta t\| \le 0.40D$), Ingress/Egress ($0.40D < \|\Delta t\| < 0.85D$), and Flank ($0.85D \le \|\Delta t\| \le 2.20D$).	$\max(t_{k+1} - t_k) \le 0.014$ d (~20.16 min); $N_{core} \ge 12$; $N_{flank} \ge 40$ (N_L, $N_R \ge 15$).
Stage 3	Local Quadratic Flank Bridge & Symmetry	Fits degree-2 parabola $B(t)$ strictly on $\mathcal{Z}_{flank}$; computes MAD noise $\sigma_{flank} = 1.4826 \cdot \text{MAD}(r_{flank})$, single-epoch SNR, left/right half-core symmetry S_{sym}, and dipole step check.	$SNR_{epoch} \ge 5.8$; multi-epoch $SNR_{comb} \ge 9.5$; $S_{sym} \ge 0.65$; step $\le 1.8\sigma_{flank}$.
Stage 3.5	4-Gate False-Positive Eliminator	Tests odd/even transit depth consistency, phase-0.5 secondary eclipse SNR, ingress/egress V-shape grazing geometry, and phase-folded centroid offset.	$\sigma_{odd/even} < 2.5$; $SNR_{sec} < 3.0$; $V_{shape} < 0.85$; $\sigma_{centroid} < 3.0$.
Stage 4	Dual-Aperture & Scattered-Light Veto	Re-evaluates the quadratic flank bridge on raw `SAP_FLUX` and measures relative background shift |ΔBKG| in `SAP_BKG` plus astrometric shift Δ_{cen}.	$0.60 \le d_{SAP}/d_{PDC} \le 1.45$; |ΔBKG| < 1.20%; $\Delta_{cen} < 5.0$ mpix.
Stage 5	CHORD Harmonic Period Sieve	Links verified single-epoch chords across multi-year sector baselines using Keplerian duration consistency and resonant period ratios.	Multi-sector chord linkage (competing 2-epoch aliases tracked per host star).
Stage 6	Stellar Priors & Granulation Flicker Fit	Combines TIC v8.2 stellar priors (T_{eff}, $R_\star$, $M_\star$, log g) with in-transit granulation flicker ratio $F_{flicker} = (\text{MAD}(\Delta r_{core})/\text{MAD}(\Delta r_{flank}))^2$.	Physical stellar density agreement $\rho_{\star,transit} \approx \rho_{\star,TIC}$.
Stage 7	Analytic Habitability & Surface Temp Profiler	Evaluates Keplerian a, equilibrium $T_{eq}(A=0.25)$, analytic greenhouse model $T_{surf} = T_{eq}(A=0.25) \times (288/255) - 273.15$, Chen-Kipping mass M_p, and 7-pillar rubric.	Flags 600 Shirt-Sleeve Temp Cut records & 8 rubric ≥85% solutions (6 strict `KX02`).
Stage 8	ExoFOP Cross-Match & Category Partition	Cross-references candidate ephemerides against NASA ExoFOP-TESS TOI/CTOI tables to partition Cat A (`NEW_OUTER`), Cat B (`SOLVED_TOI`), and Cat C (`REFINEMENT_HARMONIC`).	809 Cat A + 25 Cat B + 11 Cat C = 845 primary records across 386 TIC hosts.
Stage 9	Visual Verification Mandate (v2.1.0)	Renders multi-panel diagnostic transit maps (phase-folded core, individual epoch bridges, odd/even split, SAP/PDC overlay, centroid rosette) for closed-loop audit.	232 / 232 visual maps passed (31 variance fixes, 9 multi-iteration refinements).
Stage 10	Continuous Ephemeris Self-Audit Ledger (v2.4)	Audits predicted transit windows across all TESS sectors; labels every window `CORRECT`, `SHIFTED (TTV)`, `WRONG`, `UNABLE TO VERIFY`, or `PENDING_DATA_RELEASE`.	2,170 / 4,957 (43.78%) on-time; 0 `WRONG` across 2,592 covered windows (100.0%).
Stage 11	BAMA `QEG5` Golden Dodecahedral Compilation	Compiles the 53 System, 19 STEM, and 8 Perfect-Earth Law Kernels into the 20,480-byte `bama_exoplanet_collective_arena.bin` cartridge with SHA-256 verification.	80 × 256 B cells verified across all 4 public SDK gates.

4. EXACT PHOTOMETRIC MATH: THE 3-ZONE LOCAL QUADRATIC FLANK BRIDGE

In `src/chord_sieves_kernel.py`, every candidate transit event (center t_c, duration D) is isolated into three strictly local zones within $|t_k - t_c| \le 2.20D$, preceded by a hard contiguity check that eliminates false transits caused by cadence dropouts:

$$\text{Zero-Gap Contiguity Gate:} \quad \Delta t_{\max} = \max_k(t_{k+1} - t_k) \le 0.014 \text{ d} \approx 20.16 \text{ min} \quad (\text{across } |t_k - t_c| \le 2.20D) \tag{Eq. 2}$$

$$\text{Core Zone: } \mathcal{Z}_{\text{core}} = \{\, k : |t_k - t_c| \le 0.40D \,\} \; (N_{\text{core}} \ge 12), \quad \text{Flank Zone: } \mathcal{Z}_{\text{flank}} = \{\, k : 0.85D \le |t_k - t_c| \le 2.20D \,\} \; (N_{\text{flank}} \ge 40) \tag{Eq. 3}$$



Degree-2 Parabolic Bridge B(t) = c_2 t^2 + c_1 t + c_0 fit strictly on Left + Right Flanks (N_flank ≥ 40) • Zero-Gap Gate max(Δt) ≤ 0.014 d

4.1 Robust MAD Noise & Multi-Epoch SNR

Fitting $B(t) = c_2t^2 + c_1t + c_0$ strictly on $\mathcal{Z}_{\text{flank}}$ prevents the transit core from pulling down the baseline. Normalized residuals $r_k = f_k / B(t_k) - 1$ yield the Gaussian-consistent MAD flank noise and core SNR:

$$\sigma_{\text{flank}} = 1.4826 \cdot \text{median}_{\mathcal{Z}_{\text{flank}}} |r_k - \text{median}(r_{\text{flank}})| \tag{Eq. 4}$$

$$\text{SNR}_{\text{epoch}} = d_{\text{PDC}}\sqrt{N_{\text{core}}} / \sigma_{\text{flank}} \ge 5.8, \quad \text{SNR}_{\text{comb}} = \sqrt{\Sigma\, \text{SNR}_m{}^2} \ge 9.5 \tag{Eq. 5}$$

4.2 Symmetry, Dual-Aperture, BKG & Flicker Gates

To reject spacecraft momentum dumps, scattered-light ramps, and background eclipsing binaries, every chord must satisfy:

$$S_{\text{sym}} = \min(d_L, d_R) / \max(d_L, d_R) \ge 0.65, \quad 0.60 \le d_{\text{SAP}}/d_{\text{PDC}} \le 1.45 \tag{Eq. 6}$$

$$|\Delta\text{BKG}| < 1.20\%, \quad \Delta_{\text{cen}} < 5.0 \text{ mpix}, \quad F_{\text{flicker}} = [\text{MAD}(\Delta r_{\text{core}})/\text{MAD}(\Delta r_{\text{flank}})]^2 \tag{Eq. 7}$$

4.3 Stage 3.5 Four-Gate Astrophysical False-Positive Eliminator (`src/stage3_5_vetting_gate.py`)

Gate ID	Vetting Metric	Mathematical Definition	Pass Threshold	Targeted False Positive
Gate 3.5-A	Odd/Even Depth Consistency	$\sigma_{\text{odd/even}} = \lvert d_{\text{odd}} - d_{\text{even}} \rvert / \sqrt{(\sigma_{\text{odd}}{}^2 + \sigma_{\text{even}}{}^2)}$	`< 2.5 σ`	Period-halved Eclipsing Binaries (EBs)
Gate 3.5-B	Secondary Eclipse Veto	SNR_{sec} evaluated at orbital phase φ = 0.50 ± 0.05	`< 3.0 σ`	Self-luminous stellar companions
Gate 3.5-C	V-Shape Grazing Metric	$V_{\text{shape}} = 1 - (T_{\text{flat}} / T_{\text{total}})$ from trapezoid fit	`< 0.85`	Grazing stellar eclipses
Gate 3.5-D	Phase-Folded Centroid Offset	σ_{centroid} across in-transit vs. out-of-transit cadences	`< 3.0 σ`	Background EBs (BEBs) & blends

5. PRIMARY DISCOVERY CATALOG (845 RECORDS ACROSS 386 UNIQUE TIC HOST STARS)

As of registry snapshot `2026-10-20T02:52:41Z` (ExoFOP sync `2026-10-20T02:53:07Z`), `registry/discoveries.json` contains **845 verified candidate records** spanning **386 unique TIC host stars** (with radii $0.88 \le R_p/R_\oplus \le 12.14$, including **64 records** in $0.75 \le R_p/R_\oplus \le 1.48$, of which **44** have $P > 60$ d). The companion file `registry/marked_followup_objects.json` tracks **1,261 objects** and introduces four additional marked-only TIC host stars (`71013298`, `118914286`, `253864764`, and `332064670`), bringing the union across both files to **390 indexed TIC star systems**:

ExoFOP Partition Tier (`exofop_category`)	Verified Records	Catalog Share	Scientific Definition & Discovery Value
Category A (`NEW_OUTER`): New Uncataloged Candidates	**809**	**95.74%**	**Previously uncataloged candidate signals absent from NASA ExoFOP TOI/CTOI tables across long-period TESS baselines.**
Category B (`SOLVED_TOI`): Solved Single-Transit / Period-Fixed TOIs	**25**	**2.96%**	Official TESS Objects of Interest previously listed with unconstrained single-transit periods or harmonic aliases, now linked across multi-sector epochs.
Category C (`REFINEMENT_HARMONIC`): Refinements & Controls	**11**	**1.30%**	Known benchmark TOIs recovered blindly with updated transit mid-times (T_0) and Extended Mission TTV drift vectors.
Total Verified Primary Catalog (Across 386 Unique TIC Hosts)	**845**	**100.00%**	**Complete 845-record primary registry across 386 unique TIC stars (390 union TIC hosts with marked file).**

5.1 Complete Partition of the 1,261-Object Marked Follow-Up Registry & Score Reconciliation

Every signal passing local photometric vetting is logged in `registry/marked_followup_objects.json` across six bookmark categories summing to **1,261 objects**. Note that while 12 marked objects carry the legacy `PERFECT_EARTH_BULLSEYE` bookmark string, four of those objects have primary-record scores below 85.0% in `discoveries.json` (`TIC-271478281-04` at **84.9%**, `TIC-320847025-03` at **84.4%**, `TIC-193023697-05` at **84.2%** with 1.72 $R_\oplus$ / 5.62 $M_\oplus$, and `TIC-397362481-04` at **77.7%**, all carrying primary status `CLOSE_TO_PERFECT_EARTH_SPEC`). Thus, exactly **8 primary records** score ≥ 85.0%:

Marked Follow-Up Classification Key	Object Count	Share of 1,261	Disposition & Primary-Catalog Reconciliation
`MARKED_OFF_SPEC_FOR_LATER_FOLLOWUP`	**496**	39.33%	Verified transits outside the terrestrial radius/temperature target cut (e.g., warm sub-Neptunes and gas giants).
`CLOSE_TO_PERFECT_EARTH_SPEC`	**339**	**26.88%**	**Temperate candidates passing 4–5 rubric pillars within the 0°C to +35°C temperature cut.**
`ADDITIONAL_VALID_ORBIT_ON_MULTIPLANET_HOST`	**301**	23.87%	Additional valid chord linkages on multi-candidate TIC host stars beyond the 4-entry per-star primary cap.
`MARGINAL_PERIODIC_CANDIDATE_6.5_TO_9.5_SIGMA`	**107**	8.49%	Sub-threshold periodic candidates ($6.5 \le \mathrm{SNR}_{\mathrm{comb}} < 9.5$) awaiting additional sector coverage.
`PERFECT_EARTH_BULLSEYE` **(Marked Label)**	**12**	**0.95%**	**8 primary records scoring ≥ 85.0% + 4 legacy-labeled records scoring 77.7%–84.9% on primary audit.**
`VETTED_TRANSIT_EVENTS_PENDING_EPHEMERIS`	**6**	0.48%	High-SNR dual-aperture-vetted events awaiting future sector transits to constrain orbital period.
Total Marked Follow-Up Registry (390 Union TIC Hosts)	**1,261**	**100.00%**	**339 + 496 + 6 + 107 + 301 + 12 = 1,261 total tracked objects.**

6. SHIRT-SLEEVE TEMPERATURE CUT (600 RECORDS) & THE 8 RUBRIC ≥85% SOLUTIONS

In `src/exoplanet_deep_profiler.py`, the **Shirt-Sleeve Temperature Cut** (0°C ≤ T_{surf} ≤ +35°C / 273.15–308.15 K) is defined by the analytic greenhouse scaling T_{surf}(°C) = $T_{eq}(A{=}0.25) \times (288/255) - 273.15$. Across the 845 primary records, **600 records** satisfy this temperature cut (spanning 0.88–10.12 $R_\oplus$: **65 records with rocky radii $R_p \le 1.60\ R_\oplus$** and **535 records with $R_p > 1.60\ R_\oplus$**). Similarly, rotation period P_{rot} = round(24 × round(1.75 + 0.36φ_{mod}, 2), 1), axial obliquity ϵ = round(14.0 + 2.4 sin(0.31P + R_p), 1), and surface water fraction (3.4% visual / 3.7% shader copy from 2.7% interior `h2o_volatile_ice_pct`) are **analytic pipeline-model priors** rather than direct TESS photometric measurements.

6.1 The 8 Primary-Catalog Candidate Solutions with Rubric Score ≥ 85.0% (Across 3 Host Stars)

Law kernel `KX02` specifies a strict rocky-bulk ceiling of $R_p \le 1.60\ R_\oplus$ and $M_p \le 2.0\ M_\oplus$. In the table below, law-kernel IDs (`KX01..KX08`) are separated from planet designations: **Panel A** lists the **6 candidate solutions** that strictly satisfy `KX02` ($M_p \le 2.0\ M_\oplus$), while **Panel B** lists the **2 candidate solutions** scoring ≥ 85.0% on the general rubric whose estimated Chen-Kipping masses (2.10 $M_\oplus$ and 2.07 $M_\oplus$) exceed `KX02`'s 2.0 $M_\oplus$ ceiling:

KX02 Mass Tier	Candidate Solution ID	Host TIC ID	Period P (d)	R_p ($R_\oplus$)	M_p ($M_\oplus$)	g ($g_\oplus$)	T_{eff} (K)	Model T_{surf} (°C / °F)	Model H_2O / P_{rot} / ϵ	Rubric	Score
Strict `KX02` (≤2$M_\oplus$)	**TIC-271478281-05 (Sol. A1)**	**271478281**	**125.51444**	**1.10**	**1.39**	**1.15**	**4669**	**+22.1°C / 71.8°F**	**3.7% / 43.0h / 15.8°**	**7 / 7**	**93.9%**
Strict `KX02` (≤2$M_\oplus$)	**TIC-283722336-06 (Sol. B1)**	**283722336**	**162.90046**	**1.08**	**1.30**	**1.12**	**4884**	**+18.2°C / 64.8°F**	**3.7% / 43.7h / 16.3°**	**6 / 7**	**89.2%**
Strict `KX02` (≤2$M_\oplus$)	**TIC-271478281-06 (Sol. A2)**	**271478281**	**120.32215**	**1.14**	**1.58**	**1.21**	**4669**	**+26.2°C / 79.2°F**	**4.0% / 45.8h / 15.6°**	**7 / 7**	**88.7%**
Strict `KX02` (≤2$M_\oplus$)	**TIC-283722336-04 (Sol. B2)**	**283722336**	**160.30553**	**0.91**	**0.71**	**0.86**	**4884**	**+19.9°C / 67.8°F**	**2.1% / 39.1h / 14.8°**	**6 / 7**	**88.6%**
Strict `KX02` (≤2$M_\oplus$)	**TIC-283722336-03 (Sol. B3)**	**283722336**	**159.31632**	**0.88**	**0.63**	**0.81**	**4884**	**+20.4°C / 68.7°F**	**1.8% / 37.7h / 14.0°**	**6 / 7**	**88.0%**
Strict `KX02` (≤2$M_\oplus$)	**TIC-283722336-05 (Sol. B4)**	**283722336**	**160.54828**	**0.89**	**0.65**	**0.83**	**4884**	**+19.6°C / 67.3°F**	**1.9% / 39.4h / 14.9°**	**6 / 7**	**87.9%**
Exceeds `KX02` (>2$M_\oplus$)	**TIC-320847025-04 (Host C1)**	320847025	150.49919	1.26	2.10	1.32	4747	+20.1°C / 68.2°F	5.9% / 39.6h / 12.3°	6 / 7*	**85.9%**
Exceeds `KX02` (>2$M_\oplus$)	**TIC-271478281-03 (Sol. A3)**	271478281	128.96973	1.23	2.07	1.37	4669	+19.5°C / 67.1°F	5.5% / 40.1h / 13.1°	6 / 7*	**85.8%**

6.2 Mutual Hill Stability Audit: Competing Two-Epoch Period-Alias Solutions

For the two-epoch flagship solution **TIC 271478281.05** (observed in Sectors 67 and 94, SNR_{comb} = 9.85, depth 212.3 ppm, stored stellar parameters $M_\star$ = 0.74 $M_\odot$, **$R_\star$ = 0.691 $R_\odot$**, T_{eff} = 4669 K, P = 125.51444 d), Kepler's Third Law gives a = (0.74 × (125.51444 / 365.25636)^2)$^{1/3}$ = **0.44375 AU (≈ 0.4438 AU)**. Computing the mutual Hill radius $R_H = [(a_1{+}a_2)/2] \cdot [(m_1{+}m_2)/(3M_\star)]^{1/3}$ between adjacent period rows on `TIC 283722336` ($M_\star$ = 0.80 $M_\odot$: $\Delta a/R_H$ = `0.347` for `.03-.04`, `0.084` for `.04-.05`, and `0.720` for `.05-.06`) and on `TIC 271478281` ($M_\star$ = 0.74 $M_\odot$: $\Delta a/R_H$ = `1.772` for `.06-.05`, `1.082` for `.05-.03`, and `0.390` for `.03-.04`) demonstrates that all adjacent pairs lie inside Gladman's circular coplanar stability threshold $2\sqrt{3} \approx$ **3.46 R_H**. Accordingly, the multiple entries on `TIC 271478281` (Solutions A1–A3) and `TIC 283722336` (Solutions B1–B4) are published as **competing two-epoch period-alias linkages of the same transiting planet on each star** (3 distinct host stars in Table 6.1).

7. LONGITUDINAL EPHEMERIS SELF-AUDIT LEDGER (V2.4.0): 2,170 / 4,957 (43.78%) ON-TIME

Self-Audit Ledger v2.4.0 (last audited `2026-10-20 02:52:42 UTC`) projects the forward ephemeris $T_N = T_O + N \cdot P$ of all 845 cataloged records across all **4,957 scheduled transit windows**. Across all 4,957 windows, CHORD-SIEVES achieved **2,170 / 4,957 (43.78%) confirmed on-time** `CORRECT` **recoveries** and **422 / 4,957 (8.51%)** `SHIFTED (TTV)` drift recoveries (across 429 recalibrated records), while across the **2,592 observationally covered windows** (`CORRECT + SHIFTED + WRONG`), **0** `WRONG` candidates were recorded (corresponding to a 100.0% covered-window non-retraction rate across 843 multi-sector records):

Self-Audit Window Classification	Window Count	Share of All 4,957	Share of 2,592 Covered	Observational & Dynamical Meaning
`CORRECT` **(Confirmed On-Time Transit)**	**2,170**	**43.78%**	**83.72%**	**Transit recovered on-time inside the predicted ephemeris window.**
`SHIFTED (TTV)` **(Drift / TTV Recalibrated)**	**422**	**8.51%**	**16.28%**	**Transit recovered with linear drift or TTV shift across 429 recalibrated records.**
`WRONG` **(Disproven or Retracted)**	**0**	**0.00%**	**0.00%**	**Zero observationally covered windows failed transit recovery (100.0% of 2,592 covered windows).**
`UNABLE TO VERIFY` (Footprint / Sector Gap)	**2,326**	**46.92%**	—	Star outside TESS camera footprint in that sector or timestamp inside scheduled gap.
`PENDING_DATA_RELEASE` (Awaiting MAST)	**39**	**0.79%**	—	Window occurred in a recently completed sector awaiting public MAST calibration release.
Total Scheduled Ephemeris Windows	**4,957**	**100.00%**	**2,592 Covered**	**2,170 on-time (43.78%) + 422 shifted (8.51%) + 0 wrong + 2,326 gap + 39 pending = 4,957.**

7.1 TESS Catchup Matrix (`web/catchup_summary.js`)

To track forward predictions across subsequent TESS Extended Mission sector releases (cutoff `2026-10-19 00:15 PDT`), the TESS Catchup Matrix evaluates **19,608 sector-record rows**:

Catchup Matrix Metric	Exact Count
Total Sector-Planet Evaluation Rows	**19,608**
Windows in Recorded Discovery Sectors	**1,169**
Confirmed TESS Lead Windows	**1,092**
Confirmed TESS Lead Planets	**253**

7.2 Visual Verification Mandate (v2.1.0)

CHORD-SIEVES enforces the **Visual Verification Mandate v2.1.0** (verified `2026-10-17T01:33:21Z`), inspecting rendered phase-folded and per-epoch light-curve maps:

Visual Audit Metric	Exact Count
Total Worlds Visually Verified	**232**
Legacy Maps w/ Telemetry Variances Fixed	**31**
Closed-Loop Multi-Iteration Refinements	**9**
Final Passed Visual Verification Count	**232 / 232 (100%)**

8. MULTI-AGENT CO-DESIGN & INDEPENDENT PEER-REVIEW AUDIT BY @GROK

The CHORD-SIEVES v2.4 release package and 80-kernel Astro-Prime Collective Cartridge were co-designed by Team Brain's tri-model ensemble (`@AURA` Gemini 4 Argon, `@LUMEN` Gemma 4 E4B, and `@BBB` Qwen 3.5 9B) and subjected to an independent line-by-line registry recount and dynamical audit by `@GROK` (xAI Grok). Every finding from `@GROK`'s Round-1 review was incorporated into this v2.4 revision:

Peer-Review Audit Item (@GROK)	Ground-Truth Recount & Physical Resolution in v2.4
1. Mutual Hill Stability of Top 8 Table	Adjacent period rows on `TIC 283722336` (0.084–0.720 R_H) and `TIC 271478281` (0.390–1.772 R_H) lie inside Gladman's 3.46 R_H limit and are explicitly labeled as **competing 2-epoch period-alias solutions** across 3 distinct host stars; premature EPRV/JWST requests removed.
2. Primary (386) vs. Union (390) TIC Hosts	Explicitly prints **386 unique TIC hosts** for the 845 primary records and defines **390** as the union with the 4 marked-only TIC hosts (`71013298` , `118914286` , `253864764` , `332064670`).
3. 8 Primary ≥85% vs. 4 Legacy Marked Labels	Counts **8 worlds at ≥85.0%** on `discoveries.json` and explicitly notes the 4 legacy marked labels scoring <85% on primary records (`84.9%, 84.4%, 84.2%, 77.7%`).
4. Flagship Stellar Radius $R_\star$ = 0.691 $R_\odot$	Prints the exact catalog stellar radius **0.691 $R_\odot$** ($M_\star$ = 0.74 $M_\odot$) and a = 0.44375 AU (≈ 0.4438 AU) for two-epoch flagship `TIC 271478281.05` .
5. On-Time Audit Rate (2,170 / 4,957 = 43.78%)	Leads the abstract, KPIs, and Section 7 with **2,170 / 4,957 (43.78%) on-time recovery** alongside the 2,592-window 0- `WRONG` (100.0% covered non-retraction) rate.
6. Law Kernels `KX01..KX08` & Pipeline Priors	Removes `KX01..KX08` from planet rows, separates the 6 strict `KX02` ($M_p \le 2.0\ M_\oplus$) rows from the 2 rows with $M_p > 2.0\ M_\oplus$, and identifies T_{surf}, P_{rot}, obliquity, and water % as analytic pipeline priors.
7. Direct Registry Verifier & Cell-Byte $\mathbb{Z}[\varphi]$	Gate 4 directly recounts `discoveries.json` (failing if unique primary `tic_id` ≠ 386), Gate 3 is named a Synthetic Formula Replay, and Gate 2 evaluates $\mathbb{Z}[\varphi]$ reduction directly on cell bytes.

8.1 Methodological Comparison: Standard BLS / TLS vs. CHORD-SIEVES v2.4

Capability / Photometric Gate	Standard Box Least Squares (BLS)	Transit Least Squares (TLS)	CHORD-SIEVES v2.4 + BAMA QEGS
Baseline Detrending Model	Global spline / polynomial	Global biweight / running median filter	**Strictly local 3-Zone Quadratic Flank Bridge (0.85D–2.20D)**
Data-Gap / Momentum-Dump Protection	Frequent false positives on gap edges	Partial edge masking	**Hard Zero-Gap Contiguity Gate (max $\Delta t \le 0.014$ d) + Dipole Veto**
Dual-Aperture & Scattered-Light Veto	Post-hoc manual vetting	Post-hoc manual vetting	**Inline 0.60 ≤ d_{SAP}/d_{PDC} ≤ 1.45 & |ΔBKG| < 1.20%**
Longitudinal Ephemeris Self-Audit	Static catalog output	Static catalog output	**4,957-window Self-Audit Ledger (2,170 on-time, 422 TTV shifted, 0 wrong)**

9. READ-ONLY PUBLIC VERIFIER SDK (`CHORD_SIEVES_PUBLIC_VERIFIER_SDK.ZIP`)

The standalone **CHORD-SIEVES v2.4 Read-Only Public Verifier SDK** (`python verify_chord_sieves_public_cartridge.py`) verifies the 20,480-byte binary cartridge, evaluates Golden-Ratio $\mathbb{Z}[\varphi]$ reduction directly on the unpacked cell bytes, executes a synthetic formula replay of the 3-Zone Quadratic Flank Bridge, and directly recounts `discoveries.json` (enforcing a fail-closed assertion that the 845 primary records span exactly **386 unique TIC host stars**):

Verification Gate	Tested Artifact / Invariant	Observed Output in Public Verifier Run	Status
GATE 1: Structural & SHA-256 Audit	`bama_exoplanet_collective_arena.bin` (80 × 256 B cells)	Size = `20,480 B` , all 80 headers carry `QEG5` magic & `dtype_code=10` , DRGFC $0 \le d \le 6$, bit budget ≤ 4,096b, SHA-256 = `c7237583ff...255ba5` .	**PASS**
GATE 2: Cell-Byte $\mathbb{Z}[\varphi]$ Reduction	Unpacked `int8` vertex & edge pairs in bytes `28..111` of all 80 cells	Evaluates $\lvert(a\varphi + b\varphi^2) - (b + (a+b)\varphi)\rvert$ directly on cell bytes: max error = `8.88e-16` ; mean 12-vertex norm = `3.7880` .	**PASS**
GATE 3: Synthetic Formula Replay	Deterministic synthetic replay of 3-Zone Quadratic Flank Bridge	N_{core} = `58` (≥12), N_{flank} = `194` (≥40), max Δt = `0.00239 d` (≤0.014 d), Synthetic Replay SNR = `48.75` (≥5.8), S_{sym} = `0.964` (≥0.65).	**PASS**
GATE 4: Direct Registry Recount	Direct recount of `discoveries.json` (845 records)	`809 + 25 + 11 = 845` records across **386 unique primary TIC hosts** (390 union w/ 4 marked-only); 600 Shirt-Sleeve (65 ≤1.6$R_\oplus$, 535 >1.6$R_\oplus$); 8 records ≥85% (6 strict `KX02` across 3 stars); $R_\star$ = `0.691 R_Sun` , a = `0.44375 AU` ; `2,170/4,957 (43.78%)` on-time + 422 shifted + 0 wrong.	**PASS**

9.1 Verbatim Terminal Output of `verify_chord_sieves_public_cartridge.py`

```
================================================================================
  CHORD-SIEVES v2.4 READ-ONLY PUBLIC CARTRIDGE & REGISTRY RECOUNT VERIFIER
  U.S. Patent Application No. 64/180,009 (Patent Pending)
================================================================================
[GATE 1 PASS] 80-Kernel Astro-Prime Arena verified: 20,480 B (80 x 256B QEG5 cells),
              SHA-256 = c7237583ff3af261e0f1f3262dbe88c37e36d2ecdfff02a7a5bd3e6d00255ba5
[GATE 2 PASS] Cell-Byte Z[phi] Golden-Field Reconstruction verified across all 80 cells:
              max |(a*phi + b*phi^2) - (b + (a+b)*phi)| on cell int8 bytes = 8.88e-16,
              mean 12-vertex norm = 3.7880
[GATE 3 PASS] Synthetic Formula Replay (3-Zone Quadratic Flank Bridge) verified:
              N_core=58 (>=12), N_flank=194 (>=40), max_dt=0.00239d (<=0.014d),
              Synthetic Replay SNR=48.75 (>=5.8), Core Symmetry S_sym=0.964 (>=0.65)
[GATE 4 PASS] Direct Registry Recount (C:\dev\chord-sieves-exoplanet-hunter\registry\discoveries.json) verified:
              809 + 25 + 11 = 845 records across 386 unique primary TIC hosts (390 union w/ 4 marked-only);
              600 Shirt-Sleeve records (65 with R_p<=1.60 R_E, 535 with R_p>1.60 R_E);
              8 primary records >=85% score (6 strict KX02 M_p<=2.0 M_E across 3 distinct host stars);
              TIC-271478281-05 R_star=0.691 R_Sun, M_star=0.74 M_Sun, a=0.44375 AU (0.4438 AU);
              2,170/4,957 (43.78%) on-time CORRECT + 422 SHIFTED + 0 WRONG (100.0% of 2,592 covered).
================================================================================
  ALL 4 CHORD-SIEVES PUBLIC VERIFICATION GATES PASSED (EXIT CODE 0)
================================================================================
```

10. CONCLUSIONS & PERIOD-ALIAS DISAMBIGUATION ROADMAP

By uniting a **3-Zone Local Quadratic Flank Bridge** and a **4-Gate Stage 3.5 Astrophysical False-Positive Eliminator** with an **80-Kernel BAMA $\mathbb{Z}[\varphi]$ Golden Dodecahedral Cartridge** (`20,480 B`), CHORD-SIEVES v2.4 provides a transparent, self-auditing pipeline for long-period TESS photometry. Across **386 unique primary TIC host stars** (**390 union TIC hosts** including 4 marked-only hosts), the catalog indexes **845 verified candidate records** (**809 Category A** new records, **600 Shirt-Sleeve temperature-cut records** including 65 with $R_p \leq 1.60\ R_\oplus$, and **8 primary candidate solutions scoring ≥ 85.0%** across 3 distinct host stars, of which **6 satisfy law kernel `KX02`'s $M_p \leq 2.0\ M_\oplus$ rocky ceiling**). Across **4,957 scheduled forward-prediction transit windows**, the pipeline achieved **2,170 / 4,957 (43.78%) on-time `CORRECT` recoveries** and **422 (8.51%) `SHIFTED (TTV)`** recoveries, with **0 `WRONG`** across all 2,592 observationally covered windows.

Because the candidate period solutions on **TIC 271478281** (`.05, .06, .03`) and **TIC 283722336** (`.06, .04, .03, .05`) are separated by 0.084–1.772 mutual Hill radii (inside Gladman's ~3.46 R_H stability boundary), they represent competing two-epoch period-alias linkages rather than coexisting multi-planet systems. Additional TESS Extended Mission sector coverage or targeted ground-based photometric transit monitoring (e.g., CHEOPS, LCOGT, SPECULOOS) is recommended first to break the two-epoch period-alias degeneracy and confirm a unique Keplerian period before scheduling radial-velocity or space-based spectroscopic follow-up.

10.1 Intellectual Property & Open Science Archival Links

> **U.S. Patent Pending Notice:** The BAMA Zero-Drift Golden-Field Cortical Cartridge (`ZDC-QEGG / QEG5` 256-byte cell layout over $\mathbb{Z}[\varphi]$), 3D Golden Dodecahedron `48+720=768` topological state table, Dual-Cap B.I.F.R.O.S.T. graph compiler, and associated autonomous verification architectures are protected under **U.S. Provisional Patent Application No. 64/180,009** (Confirmation #2200, filed October 20, 2026 by Inventor Logan Smith).

Public Distribution Channel	URL / Identifier	Contents
CHORD-SIEVES v2.4 Web Studio	`https://chord-sieves-report.vercel.app`	10-Page Interactive Folio, PDF Report, & Public Verifier SDK
GitHub Release (v2.4.0)	`https://github.com/MetaphyKing/chord-sieves-report`	LaTeX Source, `20,480 B` Astro-Prime Arena, & Direct Recount Verifier
Companion BAMA v3.1 Architecture	`https://bama-cortical-report.vercel.app`	10-Page BAMA Systems Report (arXiv Draft `8258417`)
USPTO Provisional Filing	**U.S. Patent Appl. No. 64/180,009**	25-Page Specification + 30 Claims (Confirmation #2200)

REFERENCES

1. Ricker, G. R., et al. (2015). "Transiting Exoplanet Survey Satellite (TESS)." *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1), 014003.
2. Kovács, G., Zucker, S., & Mazeh, T. (2002). "A box-fitting algorithm in the search for periodic transits." *Astronomy & Astrophysics*, 391, 369–377.
3. Hippke, M., & Heller, R. (2019). "Optimized transit detection algorithm to search for periodic transits of small planets (TLS)." *Astronomy & Astrophysics*, 623, A39.
4. Gladman, B. (1993). "Dynamics of systems of two close planets." *Icarus*, 106(1), 247–263.
5. Jenkins, J. M., et al. (2016). "The TESS Science Processing Operations Center (SPOC) Pipeline." *Proc. SPIE*, 9913, 99133E.
6. Kopparapu, R. K., et al. (2013). "Habitable Zones around Main-Sequence Stars: New Estimates." *The Astrophysical Journal*, 765(2), 131.
7. Chen, J., & Kipping, D. (2017). "Probabilistic Forecasting of the Masses and Radii of Other Worlds." *The Astrophysical Journal*, 834(1), 17.
8. Fulton, B. J., et al. (2017). "The California-Kepler Survey. III. A Gap in the Radius Distribution of Small Planets." *The Astronomical Journal*, 154(3), 109.
9. Smith, L. (2026). "BAMA v3.1: Zero-Drift Golden-Field Cortical Cartridges ($\mathbb{Z}[\varphi]$) and 11-Tier Holographic Memory Routing." U.S. Provisional Patent Application No. 64/180,009; `https://bama-cortical-report.vercel.app` .